

Products of the colonic microbiota mediate the effects of diet on colon cancer risk.

It is estimated that most colon cancers can be attributed to dietary causes. We have hypothesized that diet influences the health of the colonic mucosa through interaction with the microbiota and that it is the milieu interior that regulates mucosal proliferation and therefore cancer risk. To validate this further, we compared colonic contents from healthy 50- to 65-y-old people from populations with high and low risk, specifically low risk Native Africans (cancer incidence <1:100,000; n = 17), high risk African Americans (risk 65:100,000; n = 17), and Caucasian Americans (risk 50:100,000; n = 18). Americans typically consume a high-animal protein and -fat diet, whereas Africans consume a staple diet of maize meal, rich in resistant starch and low in animal products. Following overnight fasting, rapid colonic evacuation was performed with 2 L polyethylene glycol. Total colonic evacuants were analyzed for SCFA, vitamins, nitrogen, and minerals. Total SCFA and butyrate were significantly higher in Native Africans than in both American groups. Colonic folate and biotin content, measured by *Lactobacillus rhamnosus* and *Lactobacillus plantarum* ATCC 8014 bioassay, respectively, exceeded normal daily dietary intakes. Compared with Africans, calcium and iron contents were significantly higher in Caucasian Americans and zinc content was significantly higher in African Americans, but nitrogen content did not differ among the 3 groups. In conclusion, the results support our hypothesis that the microbiota mediates the effect diet has on colon cancer risk by their generation of butyrate, folate, and biotin, molecules known to play a key role in the regulation of epithelial proliferation.